

MARTA MARIA WHEELER



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LEAD DATA SCIENTIST, CAPACITY PLANNING, FORECASTING & OPERATIONAL OPTIMIZATION

Data Scientist with 18+ years of experience building **capacity planning and forecasting systems that drive operational decisions at scale**. Specialized in predicting demand, identifying capacity risk, and translating models into actionable strategies across infrastructure and business operations. Proven track record of preventing system failures, optimizing resource allocation, and influencing cross-functional planning.

CORE EXPERTISE

Capacity Planning • Demand Forecasting • Time Series Modeling (ARIMA, Prophet, Holt-Winters) • Risk Prediction • Optimization & Decision Systems • SQL (Presto, MS SQL) • Python (Pandas, scikit-learn, statsmodels) • Experimentation • Cross-functional Leadership

COMPETENCIES

Machine Learning: Predictive Modeling, Gradient Boosting (XGBoost, LightGBM), Random Forest, Regression, Clustering, Feature Engineering

Time Series Forecasting: ARIMA, Prophet, Holt-Winters, multivariate forecasting, demand forecasting, anomaly detection, production pipeline ownership, recurring cadence delivery, forecast diagnostics & root cause analysis

Data Science & Analytics: Python (Pandas, NumPy, Scikit-learn), SQL, statistical modeling, experimentation, A/B testing

Data Platforms & MLOps: Git (GitLab, GitHub, BitBucket), CI/CD workflows, model deployment, monitoring pipelines, reproducible ML workflows

Cloud & Big Data: AWS, Spark, GCP, distributed analytics, large-scale telemetry data analysis

Visualization & Communication: Tableau, Power BI, Looker, QlickView, Unidash, data storytelling, executive reporting

Model Development and Data Delivery: End-to-End Model Development, Production Pipeline Ownership, Performance Tracking and Continuous Improvement, External Signal & Covariate Integration, Data Automation and Reporting, Data Pipelines, Advanced Excel and Analytical Tools

Business Partnership and Planning Leadership: Sales Target Alignment, Inventory and Resource Planning, Cross-Functional Collaboration, Translating Analytics into Business Decisions, Executive and Non-Technical Communication

TECHNICAL SKILLS

Python (Pandas, Scikit-learn, Sklearn, Seaborn, Numpy, Matplotlib, Joblib, Timeit, Itertools, Statsmodels, Pmdarima, FbProphet, TensorFlow Huggingface_hub, Llama_cpp, Json, PySpark), R Project, MATLAB, SAS Statistica, Mathematica, C++, MS Office, MS SQL Server, Daiquery, ORACLE, PRESTO, UNIX, Cron, PHP, HTML, CSS, JavaScript (jQuery), LaTeX, Hadoop, Spark, AWS, GCP, Rest API, Tableau, Looker, QlickView, PowerBI, Unidash Perl, CRON, JIRA, Confluence, GitHub

PROFESSIONAL EXPERIENCE

ASSOCIATE DIRECTOR, DATA SCIENCE -**Digitas**, Plano, TX

November 2025 – April 2026

- Led end-to-end automation of a cross-Line-of-Business Performance Analytics initiative, delivering dynamic reporting frameworks that improved sales visibility and marketing planning decisions across LTA, MTA, Halo, and frequency analyses.
- Led advanced analytics initiatives, statistical modeling, and machine learning efforts to solve complex marketing and business problems.
- Led end-to-end delivery of Marketing Mix Modeling, attribution, and incrementality solutions using Python, SQL, Spark, and GCP (BigQuery, Looker).
- Acted as a strategic partner to media, creative, and commerce teams, ensuring data science solutions aligned with business objectives.
- Built and scaled high-performing data science teams through mentoring, technical guidance, and cross-disciplinary collaboration.

DATA SCIENTIST – **META (via Crystal Equation)**, Irving, TX (remote)

September 2024 – June 2025

- Led end-to-end data science initiatives across infrastructure and security domains, partnering with cross-functional engineering teams to drive data-centric solutions aligned with organizational priorities (Infrastructure Capacity Planning - ARIMA models, time series forecasting; SaaS Security Maturity Measures – metric creation and presentation process, Python – Sktime, Statsmodels, Pmdarima, FbProphet).
- Translated technical and business requirements into analytical workflows, identifying and acquiring relevant datasets to support impactful insights (SQL, PRESTO).
- Developed and implemented interpretable, actionable metrics that support strategic decisions and system reliability.

- Collaborated with data engineers to optimize data pipelines and ensured data quality and governance.
- Presented findings and recommendations clearly to both technical and non-technical stakeholders, supporting leadership and strategic roadmap discussions (documents, dashboards, MS Word, Excel, PowerPoint, Unidash, Daiquery).
- Contributed to the growth of the internal data science discipline, fostering a culture of learning, knowledge sharing, and collaboration (weekly meetings, data science 'coffee talk', initiated joined data science efforts movement).

INFRASTRUCTURE PLANNING LEAD DATA SCIENTIST- CITI, Irving, TX

July 2017 – March 2024

- Demonstrated exceptional skills and expertise by leading end-to-end data engineering projects, including data acquisition, data preprocessing, and integration using SQL, Python, JavaScript (jQuery), HTML and REST APIs, MS Excel; end-to-end pipeline ownership contributed to \$20M+ in infrastructure savings in 2019 (Gold Award – MaxVM Calculator).
- Spearheaded the creation, implementation, and evaluation of advanced prediction models, including cluster breach and capacity growth prediction (kNN model: 97.58% accuracy, detected 505 of 515 actual breaches), utilizing linear and logistic regression, k-nearest neighbors (kNN), decision trees, artificial neural networks (ANN), and custom algorithms; performed root cause analysis on forecast misses and implemented model fixes to improve subsequent production cycles. R, Python (Pandas, NumPy, Joblib, Timeit, Scikit-learn, TensorFlow, Seaborn, Matplotlib).
- Successfully orchestrated the seamless migration of Data Science projects from a custom environment to an enterprise analytics system, optimizing efficiency and accessibility across the organization.
- Created and maintained relational databases for data storage and retrieval, improving data availability for cross-functional teams (MS SQL Server).
- Led the data science efforts within Citi Infrastructure Planning's Advanced Analytics, ML, and AI team, managing a tech-manager role and collaborating with a team of data scientists and analysts.
- Supported the leader's duties with a 70/30 split between technical work/training and administrative leadership, driving advanced modeling and robotic process automation for problem-solving and cost efficiencies in Citi's cloud infrastructure.
- Built predictive models using Python and R to forecast demand, optimize workload placement, and increase profitability (capacity planning, time series forecasting, Prophet model, ARIMA models); owned recurring production forecasting cadence supporting infrastructure resource allocation and pre-allocation planning decisions globally.
- Translated business objectives into AI initiatives and drove successful executions.
- Served as subject matter expert for other teams and colleagues regarding data science and AI (data science course, mentorship program).

QUANTITATIVE ANALYST - Pacific Union Financial, Irving, TX

February 2017 – July 2017

- Supervised creating and automating impactful reports, leveraging tools such as Power BI, MS SQL Server, and MS Excel to enhance the team's data visualization and decision-making capabilities.
- Developed a specialized Credit Calculator using MS Excel (VLookup Macros), providing an efficient internal tool that streamlined credit assessment processes, fostering accuracy and expediency in decision-making.
- Proficiently addressed multiple ad hoc requests with swift responsiveness, demonstrating a commitment to delivering timely solutions and ensuring operational continuity.

DATA ENGINEER - RAPP, Irving, TX

July 2016 – February 2017

- Implemented robust quality control measures, ensuring data and reports precision, streamlining an error-free database system.
- Recommended continuous improvements for data accessibility and reporting aligned with dynamic business needs.
- Meticulously worked on customer behavior, leading to the effective development of targeted marketing campaigns.
- Developed and deployed campaigns by crafting SQL queries and designing data flows using IBM Unica, optimizing data accessibility and reporting accuracy.
- Automated data processes and reports, increasing operational efficiency and reducing manual workload, showcasing a commitment to technological advancement.
- Contributed to the Data Science team's success by actively participating in creating data models using R and SQL, demonstrating a holistic approach to data-driven solutions.

DATA ANALYST / DATA SCIENTIST - Covenant Health Partners, Lubbock, TX

November 2014 – June 2016

- Automated several reports and dashboards (MS Excel - VLookup, MS Access, MS SQL Server, R – data analysis and visualization) – created reports and benchmarks to provide in-depth data analytics to support the CHP strategic objectives.
- Revolutionized reporting efficiency by expertly automating a previously time-intensive manual report, reducing the process duration from 40 hours to just 15 minutes.
- Played a significant role in database design and creation, ensuring database integrity (MS Access, MS SQL Server).
- Lead Hospital Readmissions Prevention initiative using ML methods (SQL, R).

ADDITIONAL EXPERIENCE

QUANTITATIVE ANALYST M Capital, Warsaw, Poland	2010 – 2012
VAT REPORTING SPECIALIST Hewlett-Packard Global e-Business Operations, Wroclaw, Poland	2009 – 2010
PRIVATE INSTRUCTOR Sylvan Learning Center, Wroclaw, Poland	2008 – 2009
DATA ANALYST Dialog Telecom, Wroclaw, Poland	2007 – 2008
SOFTWARE DEVELOPER / RESEARCH ASSISTANT Fraunhofer Institute for Industrial Mathematics, Kaiserslautern, Germany	2006
SOFTWARE DEVELOPER / RESEARCH ASSISTANT Fraunhofer Institute for Experimental Software, Kaiserslautern, Germany	2005 – 2006

EDUCATION

Master of Science in Mathematics – Wroclaw University of Technology, Wroclaw, Poland
European Consortium for Mathematics in Industry (ECMI) program
Mathematics for Industry and Commerce; Matlab, R, SAS (Statistical Packages subject)
Thesis: Natural Resource Price Prediction Using Time Series Methods (ARIMA, ARIMAX, ECM)

Bachelor of Science in Computer Science – Warsaw University of Technology, Warsaw, Poland
Thesis: E-learning Project for Primary Education (OCR, Artificial Neural Network)

Software Engineering – University of Kaiserslautern, Kaiserslautern, Germany
Socrates – Erasmus Scholarship, DHS Examination in German Languages

CERTIFICATIONS

<u>Applied Data Science (online course)</u> – Massachusetts Institute of Technology Online	February 2023
<u>Generative AI for NLP (online course)</u> – Great Learning Online	April 2024
Practical Data Science course – Divergence Academy, Dallas, TX	July 2018
Python Certification – Interdisciplinary Centre for Mathematical and Computational Modeling, Warsaw, Poland	January 2011

AWARDS & RECOGNITION

Gold MaxVM Calculator (Drives value for Clients)	March 2019
Copper Peggi (Delivers Results)	October 2019
Copper Scale I/O Dashboard (Champions Progress)	November 2020
Copper DMZ, Tech Refresh, Oraas dashboards (Champions Progress)	November 2020